From Points to Maps: An Ensemble Framework for Label-Efficient Remote Sensing Segmentation

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***Abstract*—High-fidelity semantic segmentation typically requires dense pixel-level annotations, which are expensive to obtain. This work investigates a robust framework for segmentation using sparse point annotations. By integrating a dual-backbone ensemble (ResNet50 and EfficientNet-B0), histogram matching for style stabilization, and Test-Time Augmentation (TTA), we bridge the gap between sparse clicks and dense maps. Our results demonstrate that a high-confidence consensus strategy combined with ensemble learning achieves a peak validation mIoU of 0.4960 using only 10 sparse points per image. This demonstrates the framework's capability to scale dense predictions from minimal human input while maintaining architectural stability.**

***Index Terms*—Semantic Segmentation, Semi-Supervised Learning, Ensemble Learning, Partial Cross Entropy, Remote Sensing.**

I. Introduction

Semantic segmentation is a cornerstone of environmental monitoring and urban planning [1]. However, the reliance on dense pixel-wise masks creates a significant data bottleneck. In projects like Landvisor, human annotators provide sparse points far more efficiently than polygons. The challenge lies in the "Incomplete Tagging" problem, where models must distinguish between truly unlabeled pixels and those to be ignored. This work utilizes an ensemble-based self-training loop to iteratively "fill in" maps from sparse anchors, ensuring style stability across diverse imagery.

II. Proposed Method

A. Ensemble Architecture and Style Stability

We utilize a dual-model ensemble consisting of two **U-Net** architectures with **ResNet50** [2] and **EfficientNet-B0** [3] encoders, both pre-trained on **ImageNet**. To ensure the stability of the picture style, **Histogram Matching** is applied during preprocessing, aligning all patches to a consistent spectral distribution to reduce variance caused by atmospheric conditions.

B. Partial Cross Entropy (pfCE)

To handle sparse supervision, we implement a Partial Cross Entropy loss. This approach follows foundational work on point-supervised segmentation [4], where the loss $\mathcal{L}_{pCE}$ is calculated strictly over the set of verified point labels Ω_p:

$$\mathcal{L}_{pCE} = -\frac{1}{|\Omega_p|} \sum_{i \in \Omega_p} \sum_{c=1}^{C} y_{i,c} \log(\hat{y}_{i,c}) \tag{1}$$

C. Consensus-Based Semi-Supervised Learning

From epoch 5 onwards, we implement a pseudo-labeling strategy based on ensemble consensus. This self-training technique utilizes the model's own high-confidence predictions as targets for unlabeled pixels [5]. Pseudo-labels $\tilde{y}$ are generated only when the averaged probability of both models exceeds threshold τ:

$$\tilde{y}_i = \begin{cases} \text{argmax}(\bar{y}_i) & \text{if } \max(\bar{y}_i) \geq \tau \\ -1 & \text{otherwise} \end{cases} \tag{2}$$

where $\bar{y}_i$ is the ensemble mean logit. Testing performance is further enhanced via **Amplification** (TTA) using horizontal flips.

III. Experiments and Results

A. Purpose and Hypothesis

We explore the interaction between label density and algorithmic consensus. We test two hypotheses:

1) **The Consensus Hypothesis:** An ensemble approach will mitigate confirmation bias more effectively than a single backbone, especially at higher thresholds ($\tau = 0.9$).
2) **The Scalability Hypothesis:** Unlike single-model baselines, the ensemble framework will show consistent gains as density increases from 1 to 10 points.

B. Experimental Process

Images were standardized to 256×256. Training was conducted for 10 epochs using the Adam optimizer ($lr = 1e-4$). We selected a 10-epoch duration to strike an optimal balance between convergence and the prevention of confirmation bias in the self-training loop. Pseudo-labeling was delayed until epoch 5 to ensure the dual-backbone ensemble established a stable baseline via pfCE before generating synthetic targets.

C. Results Analysis

The results in Table I validate the scalability of the ensemble framework. The peak performance was achieved at 10 points with a threshold of $\tau = 0.7$, reaching a Val mIoU of 0.4960 and the lowest observed Val Loss of 0.8805. We observe that while the 0.9 threshold is beneficial at very low point densities (1 point), a more inclusive 0.7 threshold becomes more effective as label density increases to 10 points. This suggests that the ensemble's consensus becomes more reliable with more anchors, allowing for a higher volume of pseudo-labels to be generated without introducing prohibitive noise.

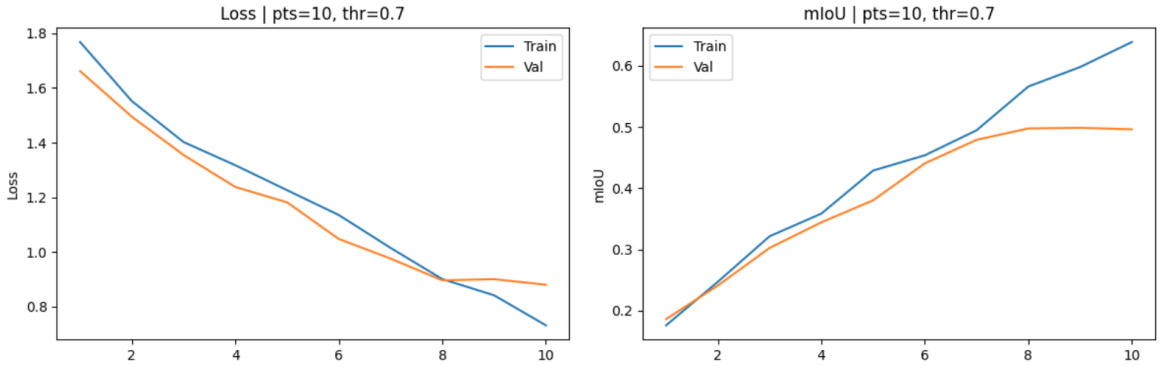

Fig. 1. Training and Validation mIoU (top) and Loss (bottom) for the optimal 10-point, $\tau = 0.9$ configuration. The curves demonstrate stable convergence within 10 epochs, justifying the early-stopping strategy to mitigate confirmation bias.

TABLE I
ENSEMBLE PERFORMANCE (10 EPOCHS): IMPACT OF DENSITY AND THRESHOLD

Points	Threshold (τ)	Train mIoU	Val mIoU	Val Loss
1	0.7	0.5317	0.3991	1.1725
1	0.9	0.5756	0.4191	1.1810
5	0.7	0.6202	0.4454	1.0456
5	0.9	0.6008	0.4826	0.9449
10	**0.7**	**0.6384**	**0.4960**	**0.8805**
10	0.9	0.5938	0.4717	0.9996

As shown in Fig. 1, the 10-epoch training duration successfully avoided the significant generalization gap seen in longer runs.

IV. CONCLUSION

This framework proves that combining ensemble learning with style-stabilized imagery allows for highly efficient segmentation. By utilizing pfCE loss and consensus-based pseudo-labeling, we successfully expanded sparse signals into dense, reliable maps.

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